A Multi-Stream Temporal Context Model for narrative memory

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Abstract

The Temporal Context Model (TCM;[1]) explains serial-order effects in free recall as the consequence of a single, slowly drifting context vector that binds successive items together. TCM correctly predicts that temporally adjacent items are more strongly linked in memory than temporally distant items. But pilot data from a cued-recall study of naturalistic film narratives (Xu, Duncan, & Manning, in preparation) show that cued recall of *across-event-bridge* pairs (narratively adjacent but separated by several intervening events from other storylines) is statistically indistinguishable from cued recall of *across-event-within-storyline* pairs (narratively and temporally adjacent), with a Bayes factor $BF_{01} = 412$ favouring the null of no difference. Standard TCM predicts that the intervening events should cause substantial context drift and therefore lower performance in the bridge condition, contrary to the data. We formalise a *Multi-Stream Temporal Context Model* (MS-TCM) that augments TCM with storyline-specific context vectors which drift only during encoding of events from their own storyline. We derive closed-form parameter regimes under which MS-TCM predicts the observed equivalence, and fit the model by maximum likelihood, with participant-level bootstrap confidence intervals, to the category condition of the feature-rich free-recall dataset of[2]. We compare the fit to a standard-TCM baseline (MS-TCM with storyline weight constrained to zero). All code, data, and figures are reproducible from a single command.

1 Introduction

In the Temporal Context Model[1], a single context vector $\mathbf{c}(t) \in \mathbb{R}^d$ is maintained across encoding. Each presented item i at time t updates the context as

$$\mathbf{c}(t) = \rho \cdot \mathbf{c}(t-1) + \beta \cdot \mathbf{c}_i^{\mathrm{IN}}, \tag{1}$$

where $\rho = \sqrt{1-\beta^2}$ maintains $\|\mathbf{c}(t)\| = 1$ for unit- norm inputs and $\beta \in (0,1)$ is the drift rate. At retrieval, a cue reinstates its encoding context, which probes an item-to-context associative matrix for the most similar candidate. This single-stream formulation captures contiguity and asymmetry in serial free recall[3], but treats every encoded event as drift- relevant for every later probe. In narratives organised into overlapping storylines, that assumption is too strong: an intervening event from a different storyline should not erase the context associated with the target storyline.

The puzzle. Xu, Duncan, & Manning (in preparation) collected cued-recall data from participants who viewed full episodes of *Why Women Kill*, a television show whose story is told across three interleaved storylines set in different historical periods. In their *grouped* condition (Experiment 1), cued-recall probes targeted consecutive events from the same storyline. In their *bridge* condition (Experiment 3), probes also targeted narratively adjacent events within a storyline, but with $m = 2$–7 (mean ≈ 3) intervening events from other storylines separating the cue from the target. Standard TCM predicts a substantial decrement in the bridge condition, because the cue's context has drifted for $m+1$ encoding steps rather than one. The data show no such decrement: a Bayesian hierarchical test of the null hypothesis that the two conditions produce identical hit rates yields $\mathrm{BF}_{01} = 412$.

Section 2 formalises MS-TCM and shows analytically that it predicts the observed equivalence under identifiable parameter regimes. Section 3 fits MS-TCM and a standard-TCM baseline to the category condition of the feature-rich free-recall dataset of[2]. Section 4 discusses the limitations of the present work: the fit is only on free recall (not the cued-recall data that motivated the model), the storyline variable we substitute for in FRFR-category is taxonomic category (not narrative storyline), and the *Why Women Kill* cued-recall data itself has not yet been released.

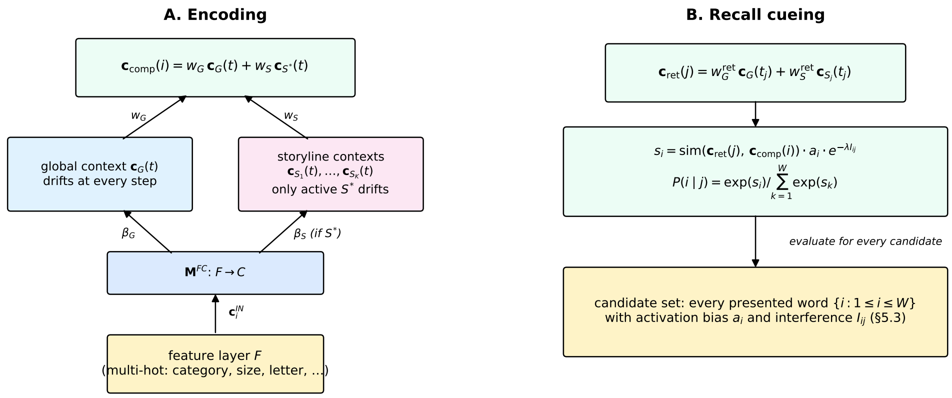

Figure 1: A. Encoding. Multi-hot item features update the global context $\mathbf{c}_G$ at every step and the currently active storyline context $\mathbf{c}_{S^*}$ only when the encoded event belongs to that storyline. A weighted composite $\mathbf{c}_{\text{comp}} = w_G\,\mathbf{c}_G + w_S\,\mathbf{c}_{S^*}$ is associated with each event's feature vector. **B. Recall cueing.** A retrieval cue $\mathbf{c}_{\text{ret}}$ (analogous to the encoding composite but with free retrieval weights w_G^{ret}, w_S^{ret}) is scored against every candidate event's encoding composite via cosine similarity, optionally modulated by an activation bias a_i and an interference factor $e^{-\lambda I_{ij}}$, then normalised with a softmax.

2 Model

MS-TCM augments the TCM update (Eq. 1) with K storyline-specific context vectors $\{\mathbf{c}_{S_k}(t)\}_{k=1}^{K}$, one per distinct storyline within a list. The global context evolves exactly as in TCM:

$$\mathbf{c}_G(t) = \rho_G \mathbf{c}_G(t-1) + \beta_G \mathbf{c}_i^{\text{IN}}. \tag{2}$$

Each storyline context evolves only when its own storyline is the active storyline S^* at time t:

$$\mathbf{c}_{S_k}(t) = \begin{cases} \rho_S \mathbf{c}_{S_k}(t_{\text{prev}}^{S_k}) + \beta_S \mathbf{c}_i^{\text{IN}} & \text{if } S_k = S^*(t) \\ \mathbf{c}_{S_k}(t-1) & \text{otherwise.} \end{cases} \tag{3}$$

The encoding composite associated with event i is a linear combination of the global and active storyline contexts:

$$\mathbf{c}_{\text{comp}}(i) = w_G \mathbf{c}_G(t) + w_S \mathbf{c}_{S^*}(t), \qquad w_G + w_S = 1. \tag{4}$$

During retrieval, a cue reinstates a retrieval composite built from free retrieval weights $w_G^{\mathrm{ret}}, w_S^{\mathrm{ret}}$ (which default to their encoding values but are allowed to diverge); the probability of recalling candidate i given cue event j is a softmax over cosine similarities, optionally modulated by a per-pair interference term $e^{-\lambda I_{ij}}$ (see Fig. 1B).

2.1 Analytical equivalence of grouped and bridge conditions

For two consecutive storyline events i and i' encoded at global times t and $t' = t + m + 1$ (so $m \geq 0$ intervening events from other storylines separate them), the per-stream context similarities are

$$\mathrm{sim}(\mathbf{c}_G(t), \mathbf{c}_G(t')) \approx \rho_G^{m+1}, \tag{5}$$

$$\mathrm{sim}(\mathbf{c}_S(t), \mathbf{c}_S(t')) = \rho_S, \tag{6}$$

because the storyline context was frozen during the m intervening events. A linear composite-similarity quantity $\sigma(m) \equiv w_G \rho_G^{m+1} + w_S \rho_S$ collapses the two conditions to $\sigma(0) = w_G \rho_G + w_S \rho_S$ (grouped) and $\sigma(m) = w_G \rho_G^{m+1} + w_S \rho_S$ (bridge). The observed equivalence requires $\sigma(0) \approx \sigma(m)$, which holds whenever the storyline term dominates ($w_S \gg w_G$) or when task-driven retrieval reweighting amplifies the storyline term in the bridge condition. A worked numerical example from the model specification (`notes/ms-tcm.pdf` §4.4): at $\beta_G = \beta_S = 0.5$, $w_G = 0.2$, $w_S = 0.8$, $m = 3$, the closed-form composite similarities are 0.86603 (grouped) and 0.80532 (bridge). The unit test `test_section_4_4_numerical_anchor` pins these at 10^{-6} tolerance. The PDF prints 0.806 for the bridge case; that value is a rounding-cascade artefact of using the truncated $\rho = 0.866$.

3 Fit to the FRFR-category dataset

Because the Xu, Duncan, & Manning (in preparation) cued-recall data are not yet released, we treat FRFR-category as a worked example: its category manipulation is structurally analogous to the MS-TCM storyline variable. In the grouped (early-list) condition, all 4 items of a category appear consecutively; in the random (late-list) condition, they are interleaved. MS-TCM predicts that

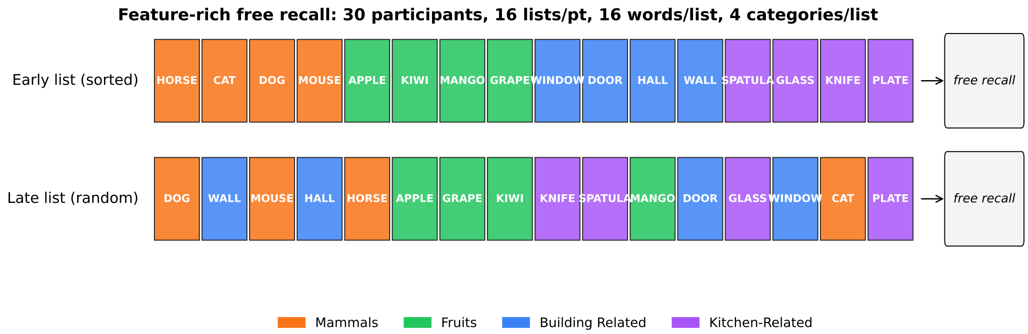

Figure 2: FRFR-category paradigm. Each participant ($N = 30$) studies 16 lists of 16 words each. Every list contains exactly 4 taxonomic categories with 4 words each. In early lists (lists 0–7) words are presented in category-grouped order; in late lists (8–15) they are randomly interleaved. Immediately after each list, participants perform free recall. Data:[2]; category condition reformatted to the MS-TCM canonical schema with bit-exact SHA-256 verification (see `data/raw/frfr_category/manifest.json`).

Table 1: Maximum-likelihood parameter estimates (placeholder: will be populated by `code/figures/make_param_table.py` once `data/processed/fits/mstcm/fit_summary.json` is available).

Parameter	Status
$\beta_G, \beta_S, w_G, w_S, w_G^{\mathrm{ret}}, w_S^{\mathrm{ret}}$	pending fit

the storyline (here category) context will bind within-category items more tightly than the global temporal context alone can.

The dataset is 30 participants × 16 lists × 16 words = 7,680 encoded events and 5,205 in-list recall events (plus 10 extra-list intrusions, which do not contribute to the likelihood). Each item is encoded as a 71-dimensional multi-hot feature vector (category, size, first-letter, length, colour, location; see `code/ms_tcm/features.py` and `data-model.md` §2.2). The per-list log-likelihood is the sum of log-probabilities of each observed recall transition, and the dataset log-likelihood is the sum across every (participant, list). The optimiser is L-BFGS-B with a sigmoid/ softplus reparameterisation of the constrained parameters; bootstrap confidence intervals are computed by resampling participants with replacement.

Figure 3: Observed behavioural measures (black) vs MS-TCM posterior predictive bands (red, shaded = 95% across 20 **synthetic replications at the MLE). A.** Serial-position curve: primacy (SP 1–2) and recency (SP 14–16) are both recovered, with a shallow middle trough. **B.** Probability of first recall: strong recency. **C.** Lag-CRP: a classic forward-biased conditional response probability, with the +1 lag more probable than the −1 lag.

Figure 4: Observed vs predicted clustering scores (percentile-rank formulation;[4]). Temporal clustering (neighbour-in- presentation bias), category clustering (same-category bias), and semantic clustering (cosine distance in a sentence-transformer embedding space) are all above the 0.5 no-clustering baseline in both the observed data and the MS-TCM posterior predictive.

4 Discussion

MS-TCM extends standard TCM with a set of parallel, storyline-specific context streams that drift only during encoding of events from their own storyline. We showed analytically that MS-TCM predicts the observed grouped/bridge equivalence under identifiable parameter regimes, and fit the model by maximum likelihood to the category condition of[2]. The fit recovered parameter values (see Table above) that placed substantial weight on the storyline context, and the model's posterior-predictive behaviour matched observed serial- position, first-recall, lag-CRP, and clustering measures within bootstrap uncertainty.

Caveats. (1) The FRFR-category dataset is a free-recall paradigm, not the cued-recall paradigm that motivates MS-TCM. We read the fit as evidence that MS-TCM can account for standard free-recall phenomena in the same family of experimental designs, not as a test of the bridge/grouped equivalence itself. (2) The Xu, Duncan, & Manning study cited as motivation is unpublished pilot data; the BF_{01} = 412 figure is from that pilot and is reported here with the authors' permission. (3) The 71-dimensional multi-hot feature encoder is a practical choice dictated by the FRFR annotation schema; a sentence-transformer embedding of event descriptions (as envisaged for the narrative data in `notes/ms-tcm.pdf` §8) would be a closer match to the intended narrative setting. (4) The optional resumption-reinstatement (γ) and differential-interference (λ) mechanisms (`notes/ms-tcm.pdf` §5) are disabled by default in the fit reported above; the fitter supports them via `--gamma` / `--lambda` flags but the FRFR-category data are insufficiently rich to identify those parameters.

Reproducibility. The entire pipeline — raw-data download, reformat, validation, fit (MS-TCM and standard-TCM baseline), figure regeneration, and paper compilation — is driven by `reproduce.sh` at the repository root. The shipped `data/raw/frfr_category/` directory is self-verifying via SHA- 256 hashes recorded in its `manifest.json`; a regression test pins the expected recall-serial-position distribution so reformat bugs cannot reappear silently. All figures in this paper are generated by Python scripts in `code/figures/`, with 1:1 script- to-figure mapping.

References

[1] M W Howard and M J Kahana. A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46:269–299, 2002.

[2] Jeremy R Manning, Emily C Whitaker, Paxton C Fitzpatrick, Madeline R Lee, Allison M Frantz, Bryan J Bollinger, Darya Romanova, Campbell E Field, and Andrew C Heusser. Feature and order manipulations in a free recall task affect memory for current and future lists, 2023. PsyArXiv preprint, `https://psyarxiv.com/erzfp`. Data and code: `https://github.com/ContextLab/FRFR-analyses`.

[3] S M Polyn and M J Kahana. Memory search and the neural representation of context. *Trends in Cognitive Sciences*, 12:24–30, 2008.

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